CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 11: Geometric Progression

Every formula and result of the chapter on one sheet

1	General G.P.	$a,\ ar,\ ar^2,\ ar^3,\ \ldots$
2	Common ratio	$r = \dfrac{t_n}{t_{n-1}}$
3	General term	$t_n = a\,r^{\,n-1}$
4	nth term from the end	$\dfrac{l}{r^{\,n-1}}$
5	Ratio of two terms	$\dfrac{t_m}{t_n} = r^{\,m-n}$
6	Sum of n terms, $r > 1$	$S_n = \dfrac{a(r^n - 1)}{r - 1}$
7	Sum of n terms, $r < 1$	$S_n = \dfrac{a(1 - r^n)}{1 - r}$
8	Sum of n terms, $r = 1$	$S_n = na$
9	Geometric mean of a and b	$\sqrt{ab}$
10	Condition for G.P.	$b^2 = ac$
11	Three terms in G.P.	$\dfrac{a}{r},\ a,\ ar$